

Between-Block Team Assignment (Blocks 1 → 2)

— Auditing the SentinelQA Store-Quality Report

EMBA 200S · Data and Decisions · Fall 2026 ·

Due: Wednesday, September 30, 11:59pm, via bCourses
Expected effort: about 1.5–2 hours of team time; the parts divide across team
Points: 100

Ground rules

You and your team are about to engage in a real-world simulation focused on what we learned in Block 1. Everything you need was covered in class, in the textbook (Ch. 13–15), or is described in this document. Please explain your answers — do not simply list numbers or formulae. We care as much about your intuition as the correctness of the answer.

One rule specific to this assignment: **finding the anomalies in this dataset earns zero points.** Your AI will locate most of them in minutes — I designed it that way, and Part A gives you the prompt. The points are for what you do afterward: saying what each finding means, what it costs, and what the company should do. You are graded on your reasoning and your recommendation.

Submission and grading

One submission per group. Put every team member’s name at the top. A complete submission is a single zip containing:

1. **One PDF:** your memo (Part E, at most one page) first, then your answers to Parts A–D, then the **red-team box** (see below), then your AI-use statement.
2. **Your workbook** (Excel or Google Sheets export) showing the computations behind your answers.

PLEASE BE BRIEF AND CONCISE WITH YOUR WRITEUP

Red-team report: a short table with three columns — *what the AI got wrong or omitted* · *what you demanded* · *where you overruled it* — with short quoted excerpts from your AI sessions.

AI-use statement (required, ungraded): a few sentences per the course policy — which tools you used and what you used them for. Transcripts not required.

Part	Topic	Points	Suggested time
A	The data	15	~15 min
B	The decline	20	~25 min
C	Confidence intervals	20	~20 min
D	The fourteen flagged stores	30	~30 min
E	Memo to the CEO	15	~25 min

Honor code

Berkeley’s honor code applies. Do not consult anyone outside your group about this assignment. Resolve every question inside the team, using any tools available to you — the textbook, your notes, statistical software, and any AI tool covered by the course AI policy. AI use is permitted and encouraged; you remain fully responsible for every number you submit. “The AI told me so” is not going to work on the final exam. All team members are expected to participate equally; teams self-manage the allocation of work.

Background

Retail and food-service chains cannot observe their own stores the way customers do, so a \$2B industry exists to do it for them: mystery-shopping firms field anonymous shoppers, score each visit against a rubric, and sell the aggregate back to the chain as a quality dashboard. The economics of the industry have two features worth knowing before you read any vendor’s report.

First, the vendor controls the entire measurement pipeline — who shops, when, where, with what rubric, and how the numbers are aggregated — and most of that pipeline is proprietary. The client sees the output, not the survey instrument, per se. Second, measurement vendors increasingly sell *remediation* alongside measurement: consulting engagements at underperforming locations, billed per store enrolled. A vendor paid per problem found may not be a neutral referee, whatever the quality of its statistics. The first question of critical appraisal — *who wrote this study, and why?* — applies to dashboards, not just journal articles (one theme of the class: economic incentives meets statistics)

Copper Kettle Coffee Co. is a 120-store regional chain. It pays SentinelQA \$480,000 a year for the Store Quality Intelligence program. In January–May, SentinelQA’s trained professional shoppers visited every store four times a month on a balanced schedule. On June 1, to cut costs and raise volume, the program switched to the “Community Shopper” platform: vetted app users who shop when and where they like, scored under a new rubric (v2). The 2027 contract is up for renewal.

Assignment

The CEO has forwarded your team SentinelQA’s Q3 Executive Briefing with one question: **“Board meets October 1. Is any of this true?”**

The briefing (attached, with its methods appendix) makes three claims: chain quality has declined “significantly” (78.1 → 74.2); fourteen named stores are statistically out of control, and their managers should be put on performance-improvement plans (PIP) with Q3 bonuses withheld; and Copper Kettle should upgrade to SentinelQA’s \$700,000 Expanded Coverage & Remediation program.

The COO wants PIPs for all fourteen managers by Friday. The CFO wants to know whether the report is even right before signing anything. Your job is to determine what, if anything, these numbers are evidence of, and to end with a recommendation.

Files and Data

File	What it is	Feeds
SentinelQA_Q3_Briefing.pdf	The vendor's report — the numbers you are checking	all parts
SentinelQA_Methods_Appendix.pdf	The vendor's data dictionary and stated methodology	Parts A, C, D
visits.csv	4,591 visit records, Jan 5–Sep 7, 2026	all parts
store_master.csv	120 stores: region, near_university, open date, renovation closures	Parts A, B

Everything is doable in Excel or Google Sheets (pivot tables, `AVERAGEIFS`, `NORM.DIST`, `T.INV.2T`) plus your AI. No Stata, R, or Python is required. These latter tools can be helpful in an agentic setup as described in class.

A note on vocabulary: the briefing calls its decline “statistically significant.” That is vocabulary from Block 2, and you do not need it here. Everything in this assignment is Block 1 material: who is in the sample, what the standard error is, what an interval prices, and where the control limits came from.

Part A — The data

[15 points, ~15 min]

Before checking any of the vendor's numbers, find out what the rows in `visits.csv` actually are. Before posting the prompt below, ask yourself - what is the aim of this specific request and what do I expect this “robot” to be doing?

Prompt (paste verbatim)

“Here are `visits.csv`, `store_master.csv`, and the vendor's methods appendix. List every column in `visits.csv` and flag anything not documented in the appendix. Then check: duplicate `visit_ids`, visits dated during a store's renovation closure, any value of `shopper_type` not listed in the appendix, and any shopper whose scoring pattern looks non-human. Report exact row counts for everything you find, and show how you got them.”

Spot-check at least two of the AI's claims yourself with a pivot table or `COUNTIF` — AIs are good at finding anomalies but can be equally fluent if/when hallucinating them.

A1 (10 pts). Make a short table — one row per finding: what it is, its exact row count, which briefing exhibit it affects, and the question you would send SentinelQA about it. There are at least five distinct findings, and one of them should change your view of “Store of the Quarter.”

A2 (3 pts). Exhibit 2 of the briefing uses `adj_score`; Exhibits 1 and 3 use `overall_score`. Work out from the data what `adj_score` is, and explain in two sentences why an adjustment with no documented recipe makes Exhibit 2 hard to audit.

A3 (2 pts). In one sentence: who computes the number that determines how many remediation engagements SentinelQA sells, and why does that matter?

Part B — Auditing the Performance Decline [20 points, ~25 min]

The headline: 78.1 in Q1, 74.2 in Q3. The arithmetic is correct. Find out what changed.

Prompt (paste verbatim)

“Using visits.csv (drop duplicate visit_ids first), build me two tables. Table 1: mean overall_score by month, separately for professional and app_volunteer shoppers. Table 2: the daypart mix (share morning / midday / evening) of each shopper program, and mean overall_score by daypart within Q3. Show row counts in every cell.”

B1 (12 pts). Is chain quality actually falling? Use the two tables to decompose the 3.9-point drop: (i) volunteers shop mostly in the evening, and evening visits score lower — reweight the Q3 daypart means back to the professional program’s balanced thirds and report how much of the drop that explains; (ii) the rest of the drop arrives on June 1, the day the shopper program changed, while the monthly trend *within* each program is flat. State your conclusion in two sentences a COO will read.

B2 (8 pts). Who is in the sample, and who is left out, after June 1? Answer for places (which stores volunteers favor), times (which dayparts), and people (who signs up to shop for an app, and how the new rubric scores). For each, one sentence on which direction it moves the measured average.

Part C — Confidence intervals

[20 points, ~20 min]

Prompt (paste verbatim)

“From visits.csv, reproduce the vendor’s headline interval: the Q3 mean of overall_score for app_volunteer visits Jul 1–Sep 7, with a 95% confidence interval — show me $\bar{x}$, s , n , and the formula. Then compute a 95% confidence interval for store S048’s Q3 mean using only S048’s own visits. Tell me which distribution you used for each interval and why.”

C1 (8 pts). Reproduce the vendor’s ± 0.4 (your unrounded number should land at ± 0.37). Then: what error does this interval price, and which of the problems you found in Parts A and B can it not see? The briefing also contains one sentence that misinterprets the interval — quote it and rewrite it correctly.

C2 (8 pts). Report S048’s interval. With $n = 4$ you must estimate the standard deviation s from the data, and thus the correct test statistic is t with $df = 3$ ($t = 3.18$), not 2 — a small sample has both a wider standard error and a larger test statistic (why?). What does this interval permit you to conclude about S048, and what doesn’t it?

C3 (4 pts). Margin of error/sample size: Using the class rule $n = 4\sigma^2/\text{MoE}^2$ with $\sigma = 6.5$: how many visits per store per quarter would give a ± 2 margin of error on a store mean? At \$40 per professional visit, what would that cost per quarter for 120 stores? One sentence comparing that with what Copper Kettle currently pays.

Part D — The fourteen flagged stores

[30 points, ~30 min]

Reconstruct the vendor's control chart first; then check it with the right standard errors.

Prompt 1 (paste verbatim)

“Reproduce SentinelQA’s control chart from visits.csv: for every store, compute the Q3 (Jul 1–Sep 7) mean of overall_score over app_volunteer visits, and flag stores below $74.2 - 2 \times (6.5/\sqrt{16}) = 70.95$. Confirm you get the same fourteen stores as the briefing, and show me each flagged store’s actual number of Q3 visits.”

Prompt 2 (paste verbatim)

“Now redo the chart with each store’s own n : flag stores below $74.2 - 2 \times (6.5/\sqrt{n_{\text{store}}})$, with $\sigma = 6.5$ known. Which of the fourteen survive? And for a healthy store with true mean 74.2 and $n = 7$ visits, compute the probability it falls below 70.95 by chance — show the z-arithmetic with NORM.DIST.”

D1 (12 pts). Build a results/verdict table for your memo — one row per flagged store: the store, your verdict (real problem / false alarm / artifact of when it was shopped), one line of evidence, and a recommended action. Use each store’s own n and standard error, plus what you learned in Parts A–B; check the flagged stores’ daypart mixes. If your own recomputation flags a store the vendor’s didn’t, add it and rule on it the same way.

D2 (8 pts). The vendor’s limit is “two standard errors” below the mean — but computed with $n = 16$ for every store. Show that for a store with $n = 7$ the limit sits only about 1.3 of its *true* standard errors below a healthy store’s expected mean, so a healthy low-volume store is flagged roughly one quarter in ten. There are 96 low-volume stores, tested every quarter. How many flags should a perfectly healthy chain expect, and what does that mean for the fourteen reported?

D3 (6 pts). State the rule you would put in the contract for next quarter, before the data arrive: your α , your limits, and your minimum visits per store (use C3). Then, before recommending that anyone’s bonus be withheld, compute the highest true quality consistent with the store’s own data ($\bar{x} + 1.645 \times 6.5/\sqrt{n}$, one-sided 95%) for S017 and for S048. For which store does the action hold up, and why is that the right standard?

D4 (4 pts). The vendor’s chart is a threshold on the quarter *average*. Name a kind of store failure this cannot catch — then ask your AI whether any store the vendor certified as healthy is trending steadily downward within the quarter. If you find one, add it to your verdict table.

Part E — Memo to the CEO

[15 points, ~25 min]

E1 (12 pts). At most one page, minimum of statistical vocabulary. It should contain: one honest sentence about the “decline”; your verdict table (as an attachment is fine) with the stores that need action now, by name; your recommendation on the bonus withholdings; and your recommendation on the contract, including at least two specific terms you would require in any renewal — for example, a documented rubric with a change log, a minimum number of visits per store, or control limits fixed in the contract before each quarter’s data arrive.

E2 (3 pts). The CEO replies: “S017 — that’s the one whose new manager started July 1. So the manager caused the drop. Fire them?” In three sentences: what would you need to know before

that claim is safe, and why can't this dataset settle it? (Block 2 begins here.)

Before you hand this in

- ☐ Names at the top; one zip per team
- ☐ Memo first (≤ 1 page), then Parts A–D, then the red-team report, then the AI-use statement